PISTAK compared with PLATON

Ring–ring parameters on 20 COD structures · 2026-11-04 · PISTAK 1c2ca85 (2026-11-04)

Summary. On the 18 structures read by PLATON, 667 ring pairs computed by PISTAK were matched to rows of PLATON *CALC ALL*. The largest difference is 0.0011 Å for the Cg···Cg distance, 0.056° for the angles α, β, γ, 0.0009 Å for the perpendicular distances and 0.0011 Å for the slippage: at most about one unit in the last digit PLATON prints (0.1° and 0.001 Å), with medians ten times smaller. These are rounding differences, not computational ones. The definitions printed by PLATON are the same as PISTAK's. The pairs PLATON lists and PISTAK does not come from how the two programs choose rings and molecule pairs, and from how PLATON names symmetry-generated atoms (section 4), not from computational errors.

1. Materials and method

Sample

From a sample of 200 CIFs drawn at random from the Crystallography Open Database (COD, public domain; series 1, 2, 4 and 7, fixed seed, list in `validation/cod_sample.txt`) we took the 20 organic, metal-free CIFs with no level-A alerts in PISTAK's quick checks, at least one stack, and the largest number of ring pairs within 6 Å. They are listed in Table 1.

COD	Space group	Rings (PLATON)	PISTAK pairs	Matched	PLATON-only classes
7212873	P 1 21/c 1	4	21	21	0
2235149	P 1 21/c 1	9	71	61	3
7236848	P -1	7	45	40	23
4032966	P -1	10	51	51	10
2222264	P 1 21/c 1	2	3	3	0
2210998	P -1	12	155	155	79
4021804	P 1 21/n 1	4	39	39	4
7229489	P -1	10	38	26	8
2227146	P 1 21/c 1	4	31	29	8
7214468	P -1	2	26	10	0
7100691	P -1	8	25	25	0
2210383	P 1 21 1	4	24	24	6
2023505	P c a 21	12	20	20	56
7119946	P -1	12	16	13	56
2233084	P -1	4	16	13	9
7159188	P 1 21/n 1	7	2	2	1
2204537	P 1 21/c 1	7	74	72	6
7159327	P 1 21/n 1	6	68	63	10
7242996		not read			
2205356		not read			
Total		124	725	667	279

Table 1. The structures compared. "PISTAK pairs": pairs of planar rings in molecules in contact, Cg···Cg ≤ 6 Å; "Matched": those found among PLATON's rows; "PLATON-only classes": PLATON pairs, counted once per symmetry class, with no counterpart in PISTAK (section 4). PLATON does not read 2205356 (the anisotropic loop precedes the atom loop) and stops at ADDSYM on 7242996.

Software

Program	Version	Use
PISTAK	1c2ca85 (2026-11-04)	pistak.core.ringpairs.ring_pairs, default thresholds
PLATON (A. L. Spek)	V-200626, built from source	CALC ALL, default settings
Python, numpy	3.13.5, 2.5.4	comparison and plots (matplotlib)

PLATON was run in text mode on each CIF, with `printf 'CALC ALL\nEXIT\n' | platon -o name.cif`; the table compared is *Analysis of Short Ring-Interactions* in `name.lis` (Cg···Cg < 6.0 Å, β < 60°; the slippage is printed only for α < 20°).

Definitions

With Cg_I, Cg_J the ring centroids, $\mathbf{n}_I$, $\mathbf{n}_J$ the normals of their mean (least-squares) planes and $\mathbf{d} = Cg_J - Cg_I$, the quantities are those of Table 2. PLATON's definitions are the ones printed in the `.lis` file; PISTAK's are in `ringpairs.py`.

Quantity	Definition	PLATON	PISTAK
Cg⋯Cg	|d|	Cg-Cg	cg_distance
α	angle between n_I and n_J (0–90°)	Alpha	alpha
β	angle between d and n_I	Beta	beta (A = I)
γ	angle between d and n_J	Gamma	gamma (B = J)
⊥ of Cg_J on plane I	|d · n_I|	CgJ_Perp	perp_a
⊥ of Cg_I on plane J	|d · n_J|	CgI_Perp	perp_b
slippage	$\sqrt{(\|d\|^2 - (d \cdot n_I)^2)}$	Slippage	slippage

Table 2. Definitions and names in the two programs. β and γ are the Janiak and PLATON angles; in PISTAK's parameter registry the same quantities for π cores are called beta and gamma.

Matching

Rings are identified by the set of their atom labels (PLATON lists them in the “n-Membered Ring (k)” lines). A PISTAK pair matches a PLATON row if the rings are the same, in either order, and the Cg⋯Cg distance differs by less than 0.006 Å; among several candidate rows (symmetry copies, a ring paired with its own image) the one closest in the angles is taken, and in the reverse order β and γ and the two perpendicular distances are swapped. The matching uses no value computed by PISTAK other than the Cg⋯Cg distance, which only selects the row.

2. Results

Quantity	Pairs	Max |Δ|	95th perc. |Δ|	Median |Δ|	PLATON rounding
Cg···Cg (Å)	667	0.0011	0.0006	0.0001	± 0.0005
α (°)	667	0.052	0.040	0.003	± 0.050
β (°)	667	0.056	0.049	0.027	± 0.050
γ (°)	667	0.050	0.045	0.024	± 0.050
CgI_Perp (Å)	667	0.0008	0.0003	0.0000	± 0.0005
CgJ_Perp (Å)	667	0.0009	0.0005	0.0001	± 0.0005
Slippage (Å)	339	0.0011	0.0007	0.0002	± 0.0005

Table 3. Differences Δ = PISTAK − PLATON over all matched pairs. The last column is half a unit in the last digit PLATON prints: a difference within it cannot be told from zero.

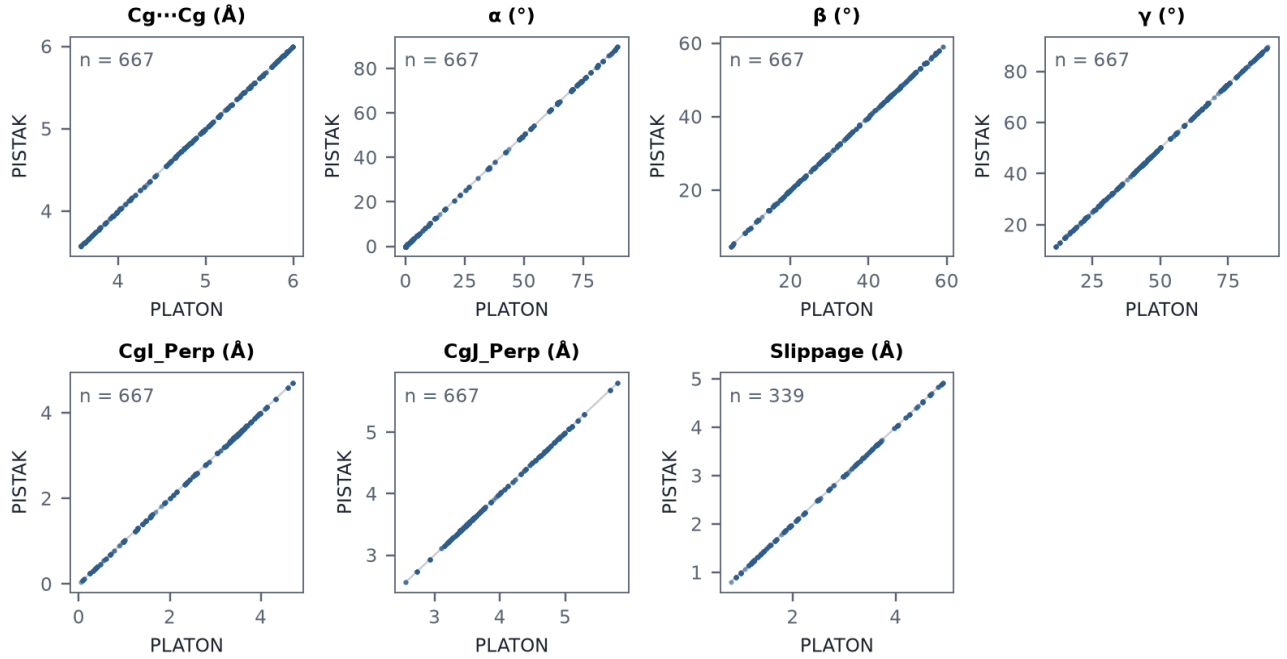


Figure 1. PISTAK values against PLATON values, pair by pair; the line is y = x.

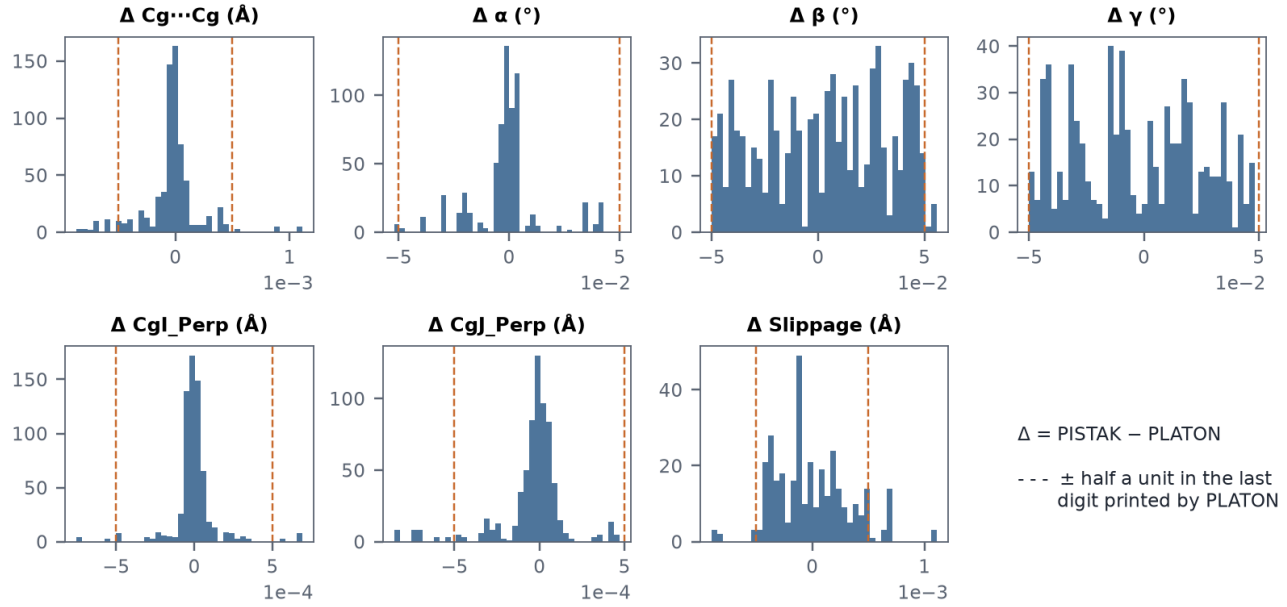


Figure 2. Distribution of the differences. The dashed lines mark PLATON's rounding.

3. Results by structure

COD	Matched	Cg···Cg (Å)	α (°)	β (°)	γ (°)	⊥ (Å)	Slippage (Å)
7212873	21	0.0007	0.029	0.041	0.041	0.0004	0.0008
2235149	61	0.0002	0.030	0.049	0.046	0.0001	0.0004
7236848	40	0.0001	0.030	0.046	0.048	0.0001	0.0004
4032966	51	0.0004	0.041	0.045	0.045	0.0004	0.0007
2222264	3	0.0001	0.030	0.039	0.039	0.0001	0.0005
2210998	155	0.0001	0.040	0.049	0.048	0.0001	0.0005
4021804	39	0.0001	0.040	0.039	0.042	0.0001	0.0003
7229489	26	0.0008	0.048	0.045	0.047	0.0008	0.0003
2227146	29	0.0004	0.030	0.050	0.050	0.0002	0.0009
7214468	10	0.0000	0.004	0.047	0.032	0.0001	0.0001
7100691	25	0.0005	0.003	0.048	0.032	0.0001	0.0002
2210383	24	0.0001	0.020	0.030	0.030	0.0001	0.0002
2023505	20	0.0003	0.040	0.052	0.048	0.0002	0.0005
7119946	13	0.0001	0.030	0.043	0.045	0.0001	0.0004
2233084	13	0.0001	0.050	0.049	0.049	0.0001	0.0004
7159188	2	0.0001	0.002	0.009	0.027	0.0001	0.0003
2204537	72	0.0011	0.052	0.056	0.045	0.0009	0.0011
7159327	63	0.0001	0.030	0.046	0.050	0.0001	0.0005

Table 4. Largest |PISTAK − PLATON| in each structure; ⊥ is the larger of CgI_Perp and CgJ_Perp.

4. Pairs PLATON lists and PISTAK does not

PLATON lists 279 classes of ring pairs (counted once per symmetry class) with no counterpart in PISTAK. None is a computational error: they come from how the two programs choose rings and pairs. The causes are in Table 5.

Cause	Explanation	Classes
Perimeter ring	8-membered rings that are the perimeter of two fused rings (5 + 5, 4 + 6). PLATON treats them as rings; PISTAK, by design, keeps only rings that are the shortest cycle through some bond.	205
Symmetry-generated atoms	PLATON names symmetry-generated atoms with a suffix (C1_a), PISTAK with the label of the original atom: label matching cannot recognise them. This is a defect of neither program.	40
Molecules not in contact	The two molecules have no heavy atoms within contact_cutoff = 4.5 Å. PLATON looks only at Cg···Cg < 6 Å; PISTAK computes the ring–ring table only for pairs of molecules in contact.	30
Disorder component	The ring contains atoms PISTAK does not have: they belong to the disorder component discarded by the major rule.	4

Table 5. Pair classes present only in PLATON, by cause. The classification is automatic, in this order: both rings exist in PISTAK (molecules not in contact); labels with a symmetry suffix; atoms missing from PISTAK (disorder); 8-membered ring (perimeter).

Conversely, PISTAK has 58 ring pairs with no PLATON row: PLATON prints only the pairs with β < 60° in at least one direction, PISTAK all those within 6 Å.

5. Interplanar distance: a definition different from PLATON's

For a pair of π cores PISTAK reports a single interplanar distance, the projection of **d** on the mean normal $\mathbf{n}_m = (n_A + n_B)/|n_A + n_B|$ (with n_B oriented like n_A). PLATON instead gives two perpendicular distances, each centroid from the other's plane. For parallel planes they coincide; Table 6 measures how far apart they are on the 1381 stack or parallel pairs of the 200 COD CIFs in the sample (whole π cores, not single rings).

Angle between planes	Pairs	median |⊥A − ⊥B|	max	median Δ	max
0–5°	1096	0.000	1.10	0.000	0.12
5–10°	132	0.539	1.94	0.007	0.36
10–15°	88	0.716	3.79	0.020	0.75
15–20°	65	1.102	4.60	0.033	0.94

Table 6. ⊥A, ⊥B: PLATON-style perpendicular distances between the cores (PISTAK parameters height_b_over_a and height_a_over_b). Δ = PISTAK interplanar distance − mean(⊥A, ⊥B). Values in Å.

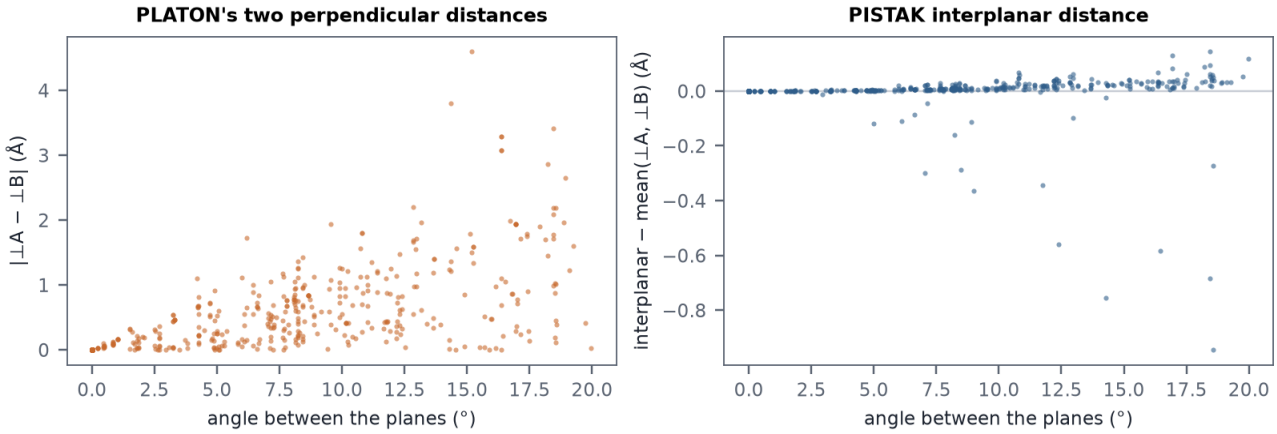


Figure 3. Left, how far apart the two perpendicular distances are as the angle between the planes grows; right, PISTAK's interplanar distance relative to their mean.

With planes tilted by 5–20° (285 pairs) the two perpendicular distances differ, in median, by 0.5 to 1.1 Å depending on the angle, and by up to 4.6 Å; the distance along the mean normal is a single number, 0.02 Å from their mean in median (at most 0.94 Å). The paper should state the definition: $d_{\perp} = |\mathbf{d} \cdot \mathbf{n}_m|$. For comparison with work that uses PLATON it is worth also reporting CgI_Perp and CgJ_Perp, which PISTAK computes identically to PLATON (section 2).

6. Limitations of the comparison

- The precision is PLATON's print precision (0.1° and 0.001 Å): the comparison shows agreement within these values, not beyond.
- Single rings are compared. The geometry of a pair of π cores (interplanar distance, long-axis and lateral offsets, overlap) has no PLATON equivalent; it is verified elsewhere in the checklist (analytical tests on hand-built crystals).
- There are 18 structures, all organic and metal-free; few have disorder.
- PLATON and PISTAK read the same CIF, but PLATON handles disorder its own way: rings in components discarded by PISTAK are not compared.

Appendix A. All matched pairs

PLATON values and difference Δ = PISTAK − PLATON for every matched pair, in PLATON's notation (ring I → ring J, “Ring” numbers of the .lis file). Δ of Cg···Cg, ⊥ and slippage in 10^{-4} Å, Δ of the angles in 10^{-2} °. Identical rows (the same ring pair found in several symmetry-equivalent molecule pairs) appear once. All 667 pairs, with PISTAK values at full precision, are in `coppie_abbinate.csv`.

COD	I→J	Cg···Cg	Δ	α	Δ	β	Δ	γ	Δ	CgI⊥	Δ	CgJ⊥	Δ	Slip.	Δ
2023505	1→1	5.040	2	0.0	−3	46.3	1	46.3	1	3.481	0	3.481	1	3.644	4
2023505	1→2	3.928	−1	1.2	0	27.0	1	27.3	−3	3.491	0	3.500	−2	1.784	−2
2023505	1→3	4.664	1	1.6	0	42.3	−4	42.9	−4	3.419	−1	3.451	0	3.137	−4
2023505	1→9	5.936	1	42.7	0	48.9	0	89.6	2	0.040	0	3.902	1	–	–
2023505	2→2	5.040	−2	0.0	−3	46.0	2	46.0	2	3.500	−1	3.500	−2	3.627	−3
2023505	2→3	4.418	1	2.5	0	39.3	1	37.7	−3	3.497	1	3.419	0	2.799	−1
2023505	2→3	5.218	0	86.3	0	20.2	−2	71.2	3	1.679	0	4.898	1	–	–
2023505	2→9	5.918	0	43.7	0	47.9	2	89.0	−4	0.107	0	3.966	1	–	–
2023505	3→2	5.290	0	86.3	0	16.4	5	74.7	−2	1.398	0	5.074	−1	–	–
2023505	3→3	5.040	−1	0.0	−2	47.7	−4	47.7	−4	3.395	−1	3.395	−1	3.725	4
2023505	3→9	5.674	0	42.4	0	46.1	0	81.0	−2	0.889	0	3.934	1	–	–
2023505	7→7	5.040	2	0.0	0	46.6	1	46.6	1	3.462	0	3.462	2	3.662	5
2023505	7→8	3.987	−3	0.8	0	29.2	5	29.4	−3	3.475	−1	3.479	−2	1.948	1
2023505	7→9	4.797	−2	1.3	0	44.1	4	44.9	2	3.396	1	3.442	−1	3.341	−2
2023505	8→8	5.040	3	0.0	−3	46.3	−1	46.3	−1	3.483	0	3.483	2	3.643	0
2023505	8→9	4.607	−1	2.0	0	42.0	3	41.2	5	3.464	0	3.422	−1	3.084	3
2023505	8→9	5.251	−1	85.8	0	19.7	3	69.7	4	1.819	−1	4.943	−2	–	–
2023505	9→3	5.669	0	80.5	0	43.3	−3	80.8	3	0.903	0	4.128	0	–	–
2023505	9→8	5.391	−1	85.8	0	15.3	2	74.3	−1	1.460	0	5.199	−2	–	–
2023505	9→9	5.040	−2	0.0	−4	47.9	−3	47.9	−3	3.381	0	3.381	−1	3.738	1
2204537	1→2	5.787	−4	5.0	1	51.4	6	46.6	−4	3.979	0	3.606	−3	4.526	−1
2204537	1→4	4.603	−2	5.4	4	44.6	1	40.2	−3	3.517	2	3.277	−2	3.232	5
2204537	1→6	4.967	4	49.2	−1	22.5	2	67.6	2	1.892	−5	4.589	−3	–	–
2204537	2→5	5.554	−9	89.3	−1	29.6	4	62.5	4	2.561	−6	4.827	−7	–	–
2204537	2→6	4.757	9	49.3	2	17.4	3	34.5	3	3.919	7	4.539	5	–	–
2204537	2→6	5.794	−6	78.3	−2	29.3	−4	77.7	−3	1.237	2	5.055	−7	–	–
2204537	4→4	3.704	1	0.0	0	21.9	1	21.9	1	3.436	6	3.437	−4	1.381	11
2204537	4→6	4.550	−7	47.9	3	8.4	3	55.8	−4	2.560	0	4.501	−9	–	–
2204537	5→6	4.720	11	48.5	−2	15.5	5	63.9	−1	2.078	0	4.548	4	–	–
2204537	5→6	5.145	−4	75.8	4	14.5	0	87.2	0	0.251	1	4.981	−3	–	–
2204537	6→1	5.762	1	74.0	1	9.2	−4	80.2	2	0.979	1	5.689	−5	–	–
2204537	6→4	4.822	−5	70.5	−5	14.5	3	74.4	−1	1.297	3	4.668	−7	–	–
2204537	6→5	4.958	5	48.5	−2	21.3	−2	67.4	1	1.904	7	4.621	−5	–	–
2210383	1→1	4.024	−1	0.0	0	30.9	1	30.9	1	3.453	0	3.453	−1	2.067	2
2210383	1→2	4.294	0	1.9	0	35.7	0	35.7	1	3.487	−1	3.487	1	2.506	−1
2210383	1→2	4.843	0	1.9	0	45.1	1	43.2	−1	3.531	0	3.418	−1	3.431	1
2210383	2→2	4.024	0	0.0	0	29.3	1	29.3	1	3.509	−1	3.509	0	1.970	2
2210383	2→3	5.774	1	54.1	0	45.8	−2	80.2	−2	0.985	0	4.027	0	–	–
2210383	2→3	5.994	−1	54.1	0	45.1	0	86.7	−3	0.348	0	4.231	−1	–	–
2210383	3→3	4.024	0	0.0	−2	27.3	−3	27.3	−3	3.577	0	3.577	0	1.844	−1
2210998	1→1	4.363	0	0.0	−3	34.9	−3	34.9	−3	3.579	0	3.579	1	2.494	3
2210998	1→2	3.619	−1	0.8	0	14.4	0	14.8	−4	3.499	0	3.505	0	0.900	2
2210998	1→3	4.690	0	3.3	0	40.3	2	43.4	0	3.408	1	3.576	0	3.035	−4
2210998	1→3	5.433	0	3.3	0	47.9	0	50.2	5	3.474	0	3.643	0	4.031	−1
2210998	1→4	3.790	0	0.8	0	22.3	2	22.7	2	3.496	1	3.506	1	1.439	1
2210998	1→4	5.884	−1	0.8	0	56.1	3	55.6	0	3.325	0	3.279	−1	4.886	−3
2210998	1→6	5.647	0	64.7	0	48.3	3	73.7	−4	1.589	0	3.755	0	–	–
2210998	2→2	4.032	0	0.0	−3	31.7	−4	31.7	−4	3.432	0	3.432	1	2.116	4
2210998	2→2	4.785	0	0.0	−3	44.7	−2	44.7	−2	3.402	0	3.402	0	3.365	−3
2210998	2→3	3.670	0	2.7	0	19.7	−1	17.0	1	3.509	0	3.455	0	1.236	5
2210998	2→3	5.443	0	2.7	0	51.6	2	49.0	−4	3.573	0	3.379	0	4.267	−2
2210998	2→4	3.694	0	0.2	0	21.8	−3	21.7	−3	3.433	0	3.431	0	1.370	−1
2210998	2→5	5.241	−1	35.4	0	52.0	4	48.7	0	3.459	0	3.224	0	–	–
2210998	2→6	5.840	0	64.2	0	54.6	−5	63.6	2	2.595	0	3.387	0	–	–
2210998	3→3	3.647	0	0.0	0	18.3	−3	18.3	−3	3.463	0	3.463	0	1.143	4
2210998	3→4	4.415	0	2.8	0	37.6	−4	40.2	−4	3.374	0	3.500	−1	2.691	1
2210998	3→4	4.665	0	2.8	0	39.8	1	42.4	−3	3.446	0	3.583	0	2.987	−4
2210998	3→6	5.865	0	61.5	0	54.3	4	83.3	0	0.684	0	3.419	0	–	–
2210998	4→4	4.557	0	0.0	−4	41.0	−1	41.0	−1	3.439	1	3.440	0	2.989	0
2210998	4→5	5.933	−1	35.2	0	57.2	1	66.7	−1	2.348	−1	3.214	0	–	–

COD	I→J	Cg⋯Cg	Δ	α	Δ	β	Δ	γ	Δ	CgI⊥	Δ	CgJ⊥	Δ	Slip.	Δ
2210998	4→6	5.280	1	64.3	0	49.5	−3	44.4	−3	3.774	0	3.431	0	–	–
2210998	4→6	5.510	0	64.3	0	52.0	4	41.5	−4	4.130	0	3.389	−1	–	–
2210998	5→5	5.378	0	0.0	−2	48.8	2	48.8	2	3.541	0	3.541	0	4.048	−1
2210998	5→6	4.834	0	80.9	0	11.7	4	85.3	−4	0.399	0	4.732	0	–	–
2210998	5→6	5.357	0	80.9	0	17.8	5	63.5	3	2.388	0	5.099	−1	–	–
2210998	6→5	5.436	0	80.9	0	26.1	2	73.3	5	1.558	0	4.881	0	–	–
2210998	6→6	4.878	0	0.0	−4	45.5	−5	45.5	−5	3.422	0	3.422	0	3.477	−5
2222264	1→1	4.326	−1	0.0	−3	22.5	0	22.5	0	3.996	−1	3.996	0	1.656	−5
2222264	1→2	3.986	1	14.3	0	20.4	2	26.0	−1	3.583	0	3.736	0	1.391	−4
2222264	2→2	3.628	−1	0.0	0	12.7	−4	12.7	−4	3.539	0	3.539	−1	0.795	1
2227146	1→1	4.089	−4	0.0	0	32.6	0	32.6	0	3.444	0	3.444	0	2.204	−9
2227146	1→2	5.250	0	53.3	0	53.1	2	40.6	−1	3.987	0	3.150	0	–	–
2227146	1→3	5.515	1	53.0	0	47.0	4	84.3	3	0.545	0	3.758	0	–	–
2227146	2→2	4.354	1	0.0	−2	35.6	−5	35.6	−5	3.543	0	3.542	1	2.532	−4
2227146	2→2	4.997	−1	0.0	−2	47.3	−2	47.3	−2	3.390	0	3.390	−1	3.671	−2
2227146	2→3	3.607	0	1.1	−1	19.3	−5	19.1	2	3.408	0	3.405	0	1.189	2
2227146	2→3	4.892	0	1.1	−1	43.8	5	44.2	4	3.504	0	3.528	0	3.389	−4
2227146	3→1	5.920	2	53.0	0	57.3	3	87.0	−4	0.314	0	3.196	−2	–	–
2227146	3→3	3.711	0	0.0	−3	23.6	−5	23.6	−5	3.401	0	3.401	1	1.483	−2
2233084	1→1	4.945	−1	0.1	−5	44.9	3	44.9	3	3.501	0	3.501	0	3.492	2
2233084	1→2	3.845	0	1.4	0	23.4	−4	22.1	−3	3.564	0	3.530	1	1.524	4
2233084	1→2	4.972	0	1.4	0	41.6	5	43.1	−3	3.632	0	3.715	−1	3.304	−1
2233084	2→2	3.776	0	0.0	−2	20.4	−5	20.4	−5	3.540	−1	3.540	0	1.313	1
2233084	2→2	3.866	0	0.0	−2	19.0	0	19.0	0	3.655	0	3.655	−1	1.259	−3
2233084	3→3	4.784	0	0.0	0	47.5	−1	47.5	−1	3.232	0	3.232	0	3.526	3
2235149	1→6	3.777	0	10.4	0	26.0	1	17.7	1	3.598	0	3.394	0	1.656	−1
2235149	2→2	5.815	0	37.9	0	50.5	−3	87.0	−2	0.306	0	3.701	−1	–	–
2235149	2→5	4.823	−2	34.7	0	43.1	1	32.0	−1	4.090	0	3.521	0	–	–
2235149	2→5	5.059	−1	72.3	0	9.7	−5	75.4	−1	1.276	0	4.987	−1	–	–
2235149	2→8	4.599	1	16.8	0	50.4	4	34.5	−3	3.792	0	2.929	1	3.546	−3
2235149	3→7	3.576	0	9.4	0	16.0	5	15.1	3	3.452	0	3.437	0	0.989	−4
2235149	3→9	3.873	0	6.2	0	25.4	2	19.3	2	3.655	0	3.498	1	1.662	3
2235149	4→8	5.972	0	81.3	0	34.0	0	87.7	−4	0.244	0	4.951	0	–	–
2235149	4→9	3.944	−1	23.0	−1	34.4	−4	11.6	−4	3.864	0	3.256	0	–	–
2235149	5→8	4.749	−1	50.6	0	43.3	3	14.8	−2	4.591	−1	3.454	0	–	–
2235149	5→8	5.089	−1	50.6	0	23.0	3	73.1	−3	1.482	−1	4.684	1	–	–
2235149	6→6	3.926	−1	0.0	−3	22.4	4	22.4	4	3.628	0	3.628	0	1.498	4
2235149	7→5	5.958	−1	87.1	0	27.5	−1	62.2	5	2.775	0	5.286	−1	–	–
2235149	8→4	5.822	−1	81.3	0	51.3	3	66.4	2	2.329	0	3.638	0	–	–
2235149	8→8	5.389	0	0.0	0	54.8	−4	54.8	−4	3.109	0	3.110	−1	4.401	3
4021804	1→3	5.559	0	83.4	0	37.7	3	75.4	−1	1.403	0	4.397	1	–	–
4021804	1→4	5.079	0	89.7	0	5.1	0	84.7	3	0.467	1	5.059	0	–	–
4021804	2→3	5.499	0	70.6	0	28.6	−4	74.4	−1	1.479	1	4.830	−1	–	–
4021804	2→4	5.149	1	60.6	0	58.0	−2	39.8	4	3.953	0	2.730	0	–	–
4021804	3→1	5.832	0	83.4	0	5.4	4	84.1	4	0.596	0	5.805	1	–	–
4021804	3→3	4.192	0	0.0	−4	27.9	−2	27.9	−2	3.706	0	3.706	0	1.960	3
4021804	3→3	4.862	0	0.0	−4	43.9	2	43.9	2	3.502	−1	3.502	0	3.372	−1
4021804	4→4	4.042	0	0.0	0	20.8	−2	20.8	−2	3.779	0	3.779	0	1.434	−1
4032966	2→2	3.938	−1	0.0	0	32.1	4	32.1	4	3.335	0	3.334	1	2.095	−3
4032966	2→6	5.892	−3	88.5	4	50.9	−2	58.8	4	3.049	0	3.717	4	–	–
4032966	2→7	5.821	4	12.5	3	53.2	−1	53.6	2	3.453	0	3.488	1	4.660	7
4032966	5→5	3.852	3	0.0	0	28.4	4	28.4	4	3.387	0	3.387	0	1.835	−1
4032966	6→3	5.816	−1	20.7	0	45.6	2	65.2	4	2.436	−3	4.067	0	–	–
4032966	8→7	5.612	−3	74.4	−2	39.6	0	67.5	−1	2.149	−2	4.324	1	–	–
7100691	2→7	3.574	1	10.2	0	19.7	−4	23.1	3	3.287	0	3.366	1	1.202	2
7100691	2→8	4.881	−5	72.7	0	23.9	−1	78.4	0	0.981	1	4.462	0	–	–
7100691	2→8	5.446	−1	72.7	0	25.8	5	79.2	2	1.018	0	4.901	0	–	–
7100691	3→7	3.739	3	16.4	0	19.2	−1	18.9	−3	3.538	0	3.532	−1	1.229	−2
7100691	3→8	4.726	−4	73.0	0	11.8	0	81.5	−1	0.699	0	4.626	0	–	–
7100691	7→2	5.875	−1	10.2	0	56.9	−5	61.6	1	2.793	0	3.212	−1	4.919	0
7100691	7→3	5.789	−3	16.4	0	51.4	−2	64.3	−1	2.511	0	3.613	−1	4.523	−1
7119946	2→7	3.572	1	4.4	0	18.4	−1	15.7	0	3.439	0	3.389	1	1.127	1
7119946	2→7	4.950	0	4.4	0	44.8	−2	47.7	3	3.330	0	3.513	0	3.487	−2
7119946	2→8	3.797	0	7.0	0	20.2	3	25.7	3	3.420	0	3.562	0	1.313	0
7119946	2→9	3.574	1	2.3	0	17.3	−2	18.7	−1	3.386	0	3.413	1	1.062	−4
7119946	2→9	4.858	0	2.3	0	47.4	−1	45.7	−3	3.395	0	3.289	0	3.575	4
7119946	7→7	4.079	0	0.0	−3	30.4	−4	30.4	−4	3.520	0	3.520	0	2.062	−3

COD	I→J	Cg⋯Cg	Δ	α	Δ	β	Δ	γ	Δ	CgI⊥	Δ	CgJ⊥	Δ	Slip.	Δ
7119946	7→8	3.616	0	2.6	0	15.9	4	16.5	1	3.467	−1	3.477	0	0.993	−1
7119946	7→9	5.635	−1	2.1	0	51.8	−4	50.2	−5	3.610	0	3.488	0	4.425	4
7119946	8→8	4.634	0	0.0	−3	40.8	0	40.8	0	3.508	0	3.508	0	3.028	1
7119946	8→9	4.025	−1	4.7	0	33.7	−2	29.3	−1	3.511	0	3.349	−1	2.232	4
7159188	1→3	3.982	−1	7.1	0	27.9	−1	28.5	3	3.499	0	3.520	−1	1.863	−3
7159327	1→2	5.808	1	76.1	0	50.8	−2	36.0	−3	4.700	0	3.672	0	–	–
7159327	1→4	5.404	0	65.0	0	35.9	2	86.1	2	0.366	1	4.376	1	–	–
7159327	1→5	4.798	1	69.9	0	4.7	1	65.2	2	2.011	1	4.782	0	–	–
7159327	2→1	5.752	0	76.1	0	35.1	1	86.3	3	0.369	1	4.706	0	–	–
7159327	2→2	5.486	0	0.0	−3	49.8	1	49.8	1	3.540	−1	3.540	−1	4.191	0
7159327	2→3	4.153	−1	2.1	0	32.6	−1	33.9	−5	3.449	0	3.499	0	2.237	0
7159327	3→1	5.164	0	74.1	0	23.4	2	72.2	3	1.576	1	4.739	1	–	–
7159327	3→3	3.726	0	0.0	−2	20.8	−1	20.8	−1	3.484	0	3.484	0	1.323	−5
7159327	4→5	4.255	−1	26.7	0	14.6	−4	40.5	−3	3.237	0	4.118	0	–	–
7159327	5→1	5.295	0	69.9	0	43.3	−3	73.8	−1	1.478	0	3.855	0	–	–
7159327	5→2	4.935	0	72.4	0	11.3	5	74.7	−3	1.304	1	4.838	0	–	–
7159327	5→4	5.912	0	10.0	0	54.8	−4	61.1	0	2.857	0	3.411	0	4.828	4
7212873	1→2	5.991	2	42.2	−2	47.9	4	43.7	4	4.329	−2	4.014	−4	–	–
7212873	1→3	4.164	−7	4.8	−1	32.3	−4	27.5	−2	3.694	−2	3.521	−1	2.223	−8
7212873	1→4	5.777	3	12.9	−1	43.5	−3	56.2	−1	3.215	0	4.193	−1	3.974	6
7212873	2→2	4.657	−6	0.0	0	35.8	2	35.8	2	3.776	−2	3.776	−2	2.725	0
7212873	2→3	4.125	3	25.3	−3	15.5	3	18.4	−4	3.915	3	3.975	1	–	–
7212873	2→4	5.652	−7	30.6	3	52.2	3	82.2	−4	0.771	−1	3.461	1	–	–
7212873	4→4	5.684	−5	0.0	0	55.8	2	55.8	2	3.193	0	3.193	0	4.702	−2
7214468	1→2	3.747	0	8.2	0	19.2	−4	13.0	3	3.650	0	3.539	0	1.230	1
7214468	1→2	3.992	0	8.2	0	27.9	−5	22.4	1	3.690	0	3.529	1	1.865	0
7229489	2→9	4.575	−1	5.7	−2	45.5	1	39.9	−5	3.512	−2	3.206	−1	3.264	−2
7229489	2→9	4.873	5	5.7	−2	48.0	4	49.0	−4	3.200	2	3.258	3	3.624	2
7229489	3→7	5.234	−3	73.0	−1	19.9	4	89.2	−1	0.075	−8	4.920	−1	–	–
7229489	4→8	5.941	3	64.5	1	34.4	0	66.7	3	2.347	0	4.902	1	–	–
7229489	6→10	3.910	−8	3.0	1	29.7	0	27.1	−1	3.481	−3	3.396	−3	1.937	−3
7229489	6→10	4.991	−2	3.0	1	46.6	1	47.0	1	3.403	3	3.429	−6	3.627	−2
7229489	7→10	5.523	1	88.1	−5	16.6	−4	88.8	2	0.114	0	5.294	1	–	–
7229489	8→4	5.498	4	64.5	1	19.4	4	72.9	−1	1.618	−1	5.185	1	–	–
7236848	1→1	4.962	0	0.0	−3	59.0	−3	59.0	−3	2.558	−1	2.558	0	4.252	3
7236848	1→2	4.137	0	2.6	0	36.7	5	34.7	2	3.401	0	3.315	0	2.475	2
7236848	1→2	5.174	−1	2.6	0	45.7	0	48.2	0	3.448	−1	3.614	0	3.703	−4
7236848	1→3	4.832	0	3.4	0	48.3	2	45.4	−3	3.394	0	3.213	0	3.609	−4
7236848	1→4	3.800	1	2.3	0	26.3	4	26.1	−1	3.413	0	3.405	0	1.686	0
7236848	1→4	3.856	−1	2.3	0	24.0	−3	25.6	4	3.476	−1	3.523	0	1.566	2
7236848	2→2	4.437	0	0.0	−2	39.0	1	39.0	1	3.448	0	3.448	0	2.793	−4
7236848	2→3	3.974	0	1.6	0	29.0	−3	30.0	−3	3.442	1	3.477	0	1.925	−4
7236848	2→3	4.647	1	1.6	0	43.5	−2	44.3	−1	3.327	0	3.372	1	3.198	−1
7236848	3→3	3.697	1	0.0	−2	25.1	−2	25.1	−2	3.349	−1	3.349	0	1.567	1
7236848	3→3	4.651	0	0.0	−2	42.7	−4	42.7	−4	3.420	0	3.420	−1	3.152	−3
7236848	3→4	5.027	1	2.3	0	46.3	2	46.3	−5	3.476	0	3.471	0	3.636	−3